

WORK SHEET - 07

d & f-BLOCK ELEMENT & METALLURGY

Single correct

- Q.1 Which series of reactions correctly represents chemical relations related to iron and its compound ?
 (A) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{FeO} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{heat}} \text{Fe}$
 (B) $\text{Fe} \xrightarrow{\text{Cl}_2, \text{heat}} \text{FeCl}_3 \xrightarrow{\text{heat, air}} \text{FeCl}_2 \xrightarrow{\text{Zn}} \text{Fe}$
 (C) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{Fe}_3\text{O}_4 \xrightarrow{\text{CO}, 600^\circ\text{C}} \text{FeO} \xrightarrow{\text{CO}, 700^\circ\text{C}} \text{Fe}$
 (D) $\text{Fe} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{H}_2\text{SO}_4, \text{O}_2} \text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\text{heat}} \text{Fe}$
- Q.2 The radius of La^{3+} is $1.06 \text{ \AA}$, which of the following given values will be closest to the radius of Lu^{3+} (At no. of Lu = 71, La = 57) -
 (A) $1.6 \text{ \AA}$ (B) $1.4 \text{ \AA}$ (C) $1.06 \text{ \AA}$ (D) $0.85 \text{ \AA}$
- Q.3 Transition elements having more tendency to form complex than representative elements (s and p-block elements) due to
 (A) availability of d-orbitals for bonding
 (B) variable oxidation states are not shown by transition elements
 (C) all electrons are paired in d-orbitals
 (D) f-orbitals are available for bonding
- Q.4 Ammonium dichromate is used in some fireworks. The green colored powder blown in the air is
 (A) CrO_3 (B) Cr_2O_3 (C) Cr (D) $\text{CrO}(\text{O}_2)$
- Q.5 Anhydrous ferric chloride is prepared by:
 (A) heating hydrated ferric chloride at a high temperature in a stream of air
 (B) heating metallic iron in a stream of dry chlorine gas
 (C) reaction of ferric oxide with HCl
 (D) reaction of metallic iron with HCl
- Q.6 Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of
 (A) nitrogen (B) oxygen (C) carbon dioxide (D) argon
- Q.7 Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statement about cerium is incorrect -
 (A) Cerium (IV) acts as an oxidising agent
 (B) The +3 oxidation state of cerium is more stable than the +4 oxidation state
 (C) The +4 oxidation state of cerium is not known in solutions
 (D) The common oxidation states of cerium are +3 and +4
- Q.8 Lanthanoid contraction is caused due to -
 (A) the same effective nuclear charge from Ce to Lu
 (B) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 (C) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 (D) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
- Q.9 Four successive members of the first row transition elements are listed below with atomic numbers.
 Which one of them is expected to have the highest $E_{\text{M}^{3+}/\text{M}^{2+}}^0$ value?
 (A) Mn ($Z = 25$) (B) Fe ($Z = 26$) (C) Co ($Z = 27$) (D) Cr ($Z = 24$)

- Q.10 The d-block element which does not exhibit difference oxidation states is
 (A) Zinc (B) Copper (C) Iron (D) Manganese
- Q.11 When MnO_2 is fused with KOH in air it gives
 (A) K_2MnO_4 (B) KMnO_4 (C) Mn(OH)_2 (D) Mn_3O_4
- Q.12 When chromite ore is fused with NaOH in the presence of air, the product formed is
 (A) $\text{Na}_2\text{Cr}_2\text{O}_7$ (B) Cr_2O_3 (C) Na_2CrO_4 (D) $\text{K}_2\text{Cr}_2\text{O}_7$
- Q.13 Which of the following does not contain Mg:
 (A) magnetite (B) magnesite (C) asbestos (D) carnallite
- Q.14 Refining of silver is done by:
 (A) liquation (B) poling (C) cupellation (D) van Arkel method
- Q.15 In the extraction of nickel by Mond process, the metal is obtained by:
 (A) electrochemical reduction (B) thermal decomposition
 (C) chemical reduction by aluminium (D) reduction by carbon
- Q.16 Refractory materials are generally used in furnaces because
 (A) they are chemically inert (B) they can withstand high temperature
 (C) they do not contain impurities (D) they decrease melting point of ore
- Q.17 When an impurity in a metal has greater affinity for oxygen and is more easily oxidises than the metal itself. Then, the metal is refined by
 (A) cupellation (B) zone-refining (C) distillation (D) electrolytic process
- Q.18 In the aluminothermite process, Al acts as
 (A) An oxidising agent (B) A flux (C) A reducing agent (D) A solder
- Q.19 The method of zone refining of metals is based on the principle of:
 (A) Greater mobility of the pure metal than that of impurity.
 (B) Higher melting point of the impurity than that of the pure metal.
 (C) Greater noble character of the solid metal than that of the impurity
 (D) Greater solubility of the impurity in the molten state than in the solid
- Q.20 The chemical composition of "slag" formed during the smelting process in the extraction of copper is:
 (A) $\text{Cu}_2\text{O} + \text{FeS}$ (B) FeSiO_3 (C) CuFeS_2 (D) $\text{Cu}_2\text{S} + \text{FeO}$
- Q.21 In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are:
 (A) O_2 and CO respectively (B) O_2 and Zn dust respectively
 (C) HNO_3 and Zn dust respectively (D) HNO_3 and CO respectively
- Q.22 Sulfide ores are common for the metals :
 (A) Ag, Cu and Pb (B) Ag, Cu and Sn
 (C) Ag, Mg and Pb (D) Al, Cu and Pb
- Q.23 $\text{Cr}_2\text{O}_7^{2-} \xrightleftharpoons[\text{Y}]{\text{X}} 2\text{CrO}_4^{2-}$, X and Y are respectively
 (A) $\text{X} = \text{OH}^-$, $\text{Y} = \text{H}^+$ (B) $\text{X} = \text{H}^+$, $\text{Y} = \text{OH}^-$
 (C) $\text{X} = \text{OH}^-$, $\text{Y} = \text{H}_2\text{O}_2$ (D) $\text{X} = \text{H}_2\text{O}_2$, $\text{Y} = \text{OH}^-$

Comprehension

Paragraph for question nos. 24 to 27

Some of the lower quality ore are mixed with Haemite for Fe-extraction like Siderite, Limonite and magnetite. Roasted ores are mixed with coke and lime stone, finally heated in the blast furance to get molten Fe.

- Q.24 Choose the correct statement regarding the roasting process here.
(A) It is illogical because there is no sulphide ore.
(B) Limonite is converted into magnetite during roasting
(C) All FeO are converted into Fe_2O_3 .
(D) None of these
- Q.25 Which of the following reaction is least favoured at the hearth.
(A) $\text{SiO}_2 + 2\text{C} \longrightarrow \text{Si} + 2\text{CO}$
(B) $\text{SiO}_2 + \text{C} \longrightarrow \text{Si} + \text{CO}_2$
(C) $\text{Mn}_3\text{O}_4 + 4\text{C} \longrightarrow 3\text{Mn} + 4\text{CO}$
(D) $\text{Ca}_3(\text{PO}_4)_2 + 3\text{SiO}_2 \longrightarrow 3\text{CaSiO}_3 + \text{P}_2\text{O}_5$
- Q.26 Which of following gas is not present in blast furnace gas.
(A) O_2 (B) CO_2 (C) CO (D) N_2
- Q.27 Which of following steel treatment process is applicable for softening of the steel.
(A) Leaching (B) Quenching (C) Nitriding (D) Annealing

Paragraph for question nos. 28 to 31

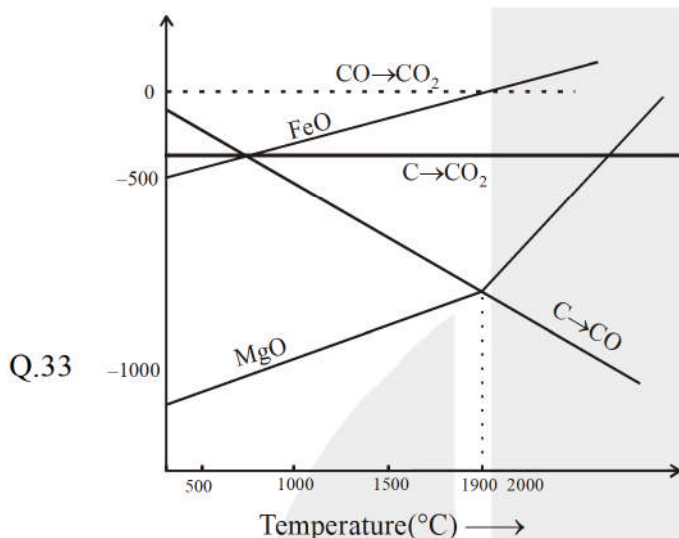
Beneficiation of the ore can be done by different ways. Which is increasing the concentration of ore removing impurities. While refining is the process by which concentration of metal is increased in the final product.

- Q.28 Which of the following compound will be reduced by self reduction.
(A) PbS (B) HgS (C) Cu_2S (D) All
- Q.29 The liquation process is applicable for
(A) For high melting metals (B) For low melting metals
(C) (A) and (B) both (D) None of these
- Q.30 What is the correct match for the name of the ore in column I and formula in column II
- | Name of the ore | Formula |
|---------------------|--|
| (A) Zincite | $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ |
| (B) Malachite green | $\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$ |
| (C) Anglesite | PbCO_3 |
| (D) Magnetite | MgCO_3 |
- Q.31 The basic principle to apply vapour phase refining is/are
(A) The intermediate compound has to be volatile
(B) The intermediate compound has to be relatively thermally unstable.
(C) (A) and (B) both
(D) Neither (A) nor (B)

Paragraph for question nos. 32 to 34 Most metals are obtained from minerals. A mineral is obtained by mining and is a naturally occurring substance with a range of chemical composition.

Q.32 In self reduction process of metal sulphide which of the following gas is evolved.

- (A) CO_2 (B) CO (C) SO_2 (D) NO_2



Using the Ellingham diagram, show the lowest temperature at which MgO can be reduced to metal by carbon.

- (A) 1500°C (B) 1120°C (C) 1900°C (D) 1600°C

Q.34 Choose the **incorrect** option :

- (A) Fractional distillation is used when there is difference in densities of ore and gangue.
 (B) Mond's process is used for purification of Ni.
 (C) Matte is ($\text{Cu}_2\text{S} + \text{FeS}$)
 (D) Heating of mild steel in presence of NH_3 is called nitriding.

More than one may be correct

Q.35 Common impurities present in Bauxite are

- (A) CuO (B) ZnO (C) Fe_2O_3 (D) SiO_2

Q.36 The correct statement(s) about transition elements is/are

- (A) the most stable oxidation state is +3 and its stability decreases across the period
 (B) transition elements of 3d-series have almost same atomic sizes from Cr to Cu
 (C) the stability of +2 oxidation state increases across the period
 (D) some transition elements like Ni, Fe, Cr may show zero oxidation state in some of their compounds

Q.37 Which of the following statements concern with transition metals?

- (A) compounds containing ions of transition elements are usually coloured
 (B) the most common oxidation state is +3
 (C) they show variable oxidation states, which differ by two units only
 (D) they easily form complexes

Q.38 Roasting can be performed in

- (A) blast furnace (B) reverberatory furnace
 (C) Bessemer's converter (D) electric furnace

Q.39 Amphoteric oxide(s) is/are

- (A) Al_2O_3 (B) SnO (C) ZnO (D) Fe_2O_3

- Q.40 Acidified KMnO_4 can be decolourised by
 (A) SO_2 (B) H_2O_2 (C) FeSO_4 (D) $\text{Fe}_2(\text{SO}_4)_3$
- Q.41 Ion(s) having non zero magnetic moment (spin only) is/are
 (A) Sc^{3+} (B) Ti^{3+} (C) Cu^{2+} (D) Zn^{2+}
- Q.42 Which of the following are true for electrolytic extraction of aluminium
 (A) cathode material contains graphite (B) anode material contains graphite
 (C) cathode reacts away forming CO_2 (D) anode reacts away forming CO_2
- Q.43 Auto reduction process is used in extraction of
 (A) Cu (B) Hg (C) Al (D) Fe
- Q.44 Which of the following reduction reactions are actually employed in commercial extraction of metals?
 (A) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$
 (B) $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
 (C) $2\text{Na}[\text{Au}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Au}$
 (D) $\text{Cu}_2\text{S} + \text{Pb} \rightarrow \text{Cu} + \text{PbS} \downarrow$
- Q.45 During estimation of oxalic acid Vs KMnO_4 , self indicator is
 (A) KMnO_4 (B) oxalic acid (C) K_2SO_4 (D) MnSO_4
- Q.46 The chemical process in the production of steel from haematite ore involve:
 (A) reduction (B) oxidation
 (C) reduction followed by oxidation (D) oxidation followed by reduction
- Q.47 Upon heating with Cu_2S , the reagent(s) that give copper metal is/are
 (A) CuFeS_2 (B) CuO
 (C) Cu_2O (D) CuSO_4
- Q.48 Hoop's process of purification of aluminium involves formation of layers during electrolysis. It involves
 (A) the three layers have same densities but different materials.
 (B) the three layers have different densities
 (C) the upper layer is of pure aluminium which acts as a cathode
 (D) the bottom layer is of impure aluminium which acts as an anode and middle layer consists of cryolite and BaF_2 .
- Q.49 Calcium silicate slag formed in extraction of iron
 (A) prevents the reoxidation of molten iron.
 (B) catalyses the combustion of carbon.
 (C) reduces CO_2 to CO at the bottom of the furnace.
 (D) is used in cement industry.

Match the column

Q.50	Column-I	Column-II
(A)	Highest density	(P) Os
(B)	Colourless salts	(Q) Cr
(C)	Maximum magnetic moment	(R) Zn
(D)	Variable oxidation state	(S) Mn

Q.51 Match the extraction processes listed in column I with metals listed in column II.

Column I	Column II
(A) Self reduction	(P) Lead
(B) Carbon reduction	(Q) Silver
(C) Complex formation and displacement by metal	(R) Copper
(D) Decomposition of iodide	(S) Boron

Q.52 Match the conversions in **Column I** with the type(s) of reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	Column II
(A) $\text{PbS} \rightarrow \text{PbO}$	(P) Roasting
(B) $\text{CaCO}_3 \rightarrow \text{CaO}$	(Q) Calcination
(C) $\text{ZnS} \rightarrow \text{Zn}$	(R) Carbon reduction
(D) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$	(S) Self reduction

Subjective

Q.53 In order to concentrate galena (which contains ZnS as impurity) by froth-floatation process, sodium cyanide is used as depressant. NaCN dissolves ZnS due to formation of water soluble complex(A). Find 'wxyz',

where w = coordination number of central metal ion in complex ion of (A)
 x = number of unpaired electrons in (A).
 y = total number of possible linkage isomers of (A) including (A).
 z = maximum number of atoms in a single plane in the complex ion of (A).

Q.54 Find the number of elements which are extracted by carbon reduction method.
 Fe, Sn, Zn, Pb, Na, Mn, Au, Ag

Q.55 How many of the following metals is extracted by self reduction process?
 Pb, Au, Mg, Sn, Fe, Al, Ag, Hg, Na, Zn, Cu, Ca
If your answer is 2, write the answer as 0002.

ANSWER KEY

Q.1 C	Q.2 D	Q.3 A	Q.4 B	Q.5 B
Q.6 B	Q.7 C	Q.8 B	Q.9 C	Q.10 A
Q.11 A	Q.12 C	Q.13 A	Q.14 C	Q.15 B
Q.16 B	Q.17 A	Q.18 C	Q.19 D	Q.20 B
Q.21 B	Q.22 A	Q.23 A	Q.24 C	Q.25 B
Q.26 A	Q.27 D	Q.28 D	Q.29 B	Q.30 B
Q.31 C	Q.32 C	Q.33 C	Q.34 A	Q.35 CD
Q.36 ABCD	Q.37 ABD	Q.38 AB	Q.39 ABC	Q.40 ABC
Q.41 BC	Q.42 ABD	Q.43 AB	Q.44 BC	Q.45 AD
Q.46 C	Q.47 BCD	Q.48 BCD	Q.49 AD	
Q.50 (A) P (B) R (C) Q (D) P,Q,S	Q.51 (A) P, R ; (B) P ; (C) Q ; (D) S			
Q.52 (A) P; (B) Q; (C) P,R (D) P,S	Q.53 4055	Q.54 0005	Q.55 0003	

